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Artificial intelligence in process safety: A review of opportunities, challenges, and future directions for the chemical process industries

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ABSTRACT

The Chemical Process Industries (CPIs) are undergoing a data-driven transformation, as artificial intelligence (AI), the Industrial Internet of Things (IIoT), and advanced analytics are redefining critical safety operations. This review consolidates recent applications of AI for hazard identification, dynamic risk assessment, early incident detection, and proactive barrier management in refineries and petrochemical facilities, emphasizing reported enhancements of 30–60 % in anomaly detection and near-miss classification compared to traditional methodologies. Ongoing challenges include sparse and biased incident datasets, limited interpretability, difficulties in validating algorithms under actual plant conditions, and regulatory ambiguities that limit widespread deployment. To mitigate these deficiencies, the paper proposes a multilayered framework that aligns AI functions with process safety management (PSM) elements, including asset integrity, alarm management, and emergency response. Future research priorities include the development of explainable, certifiable artificial intelligence, digital-twin-driven predictive risk assessment, and the integration of functional safety and risk governance standards, such as International Electrotechnical Commission (IEC) 61511 and ISO 31000. The review delineates pathways to establish safer, cleaner, and more resilient chemical and energy systems globally.

1. Introduction

Process safety within the CPIs is an essential element of industrial risk management, emphasizing the prevention of catastrophic events such as explosions, fires, and hazardous emissions that pose significant threats to human life, property, and environmental integrity (Soltanzadeh et al., 2024). Notable catastrophic events, including Bhopal (1984), Texas City (2005), and Tianjin (2015), serve as enduring exemplars underscoring the critical necessity for comprehensive safety management systems. Over the last four decades, systematic approaches including Hazard and Operability Studies (HAZOP), Layers of Protection Analysis (LOPA), and Quantitative Risk Assessment (QRA) have established rigorous frameworks for identifying, evaluating, and mitigating process-related hazards (Oboh, 2023). Despite their proven effectiveness, these traditional methodologies often rely on expert judgment, static assumptions, and discrete-time analyses, which limit their operational efficacy in dynamic plant environments characterized by rich, continuously shifting data streams (Tiwari and Malluri, 2023).

Modern chemical manufacturing facilities generate substantial volumes of process, maintenance, and environmental data via distributed control systems (DCS), safety instrumented systems (SIS), and IIoT sensor networks. These extensive and complex datasets present both

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opportunities and challenges for managing process safety (Abedsoltan and Abedsoltan, 2024). On the one hand, they encompass early indicators of abnormal operational states and the latent degradation of safety barriers; on the other hand, human operators and conventional analytical tools often struggle to interpret such high-dimensional data in real time. This evolving landscape has catalyzed the integration of AI into process safety frameworks to enhance human decision-making and improve the responsiveness of safety systems (Rajurkar, 2024).

AI, comprising Machine learning (ML), Deep learning (DL), and digital twin technologies, offers capabilities for pattern recognition, anomaly detection, and predictive risk assessment that can fortify existing safety systems. AI-driven analytical techniques have shown promise in early fault detection, dynamic risk assessment, incident prediction, and human-machine decision support across refineries, petrochemical facilities, and specialty-chemical production environments. Simultaneously, the implementation of AI in high-hazard settings introduces novel technical and regulatory challenges, such as model explainability, data integrity, cybersecurity, and the validation of AI functionality under uncertain, non-stationary, or infrequent-event conditions. These challenges highlight the imperative of aligning AI applications with established principles of process safety, functional safety standards (e.g., IEC 61511), and risk governance frameworks (e.g., ISO 31000, CCPS) (Park and Kang, 2024).

Recent academic literature has investigated various dimensions of digital transformation in process industries, including predictive maintenance, asset integrity management, alarm rationalization, and human factors analytics (Zamora-Pastor et al., 2025). Nevertheless, a thorough evaluation of how AI technologies directly facilitate the prevention of losses and the mitigation of risks, particularly with respect to major accident hazards (MAHs), barrier management, and integration into the safety lifecycle, remains fragmented (Khurram et al., 2025). Numerous existing studies emphasize performance enhancements or computational efficiency of AI models without systematically relating these to process safety functions, levels of autonomy, or assurance requirements throughout the safety lifecycle.

The existing literature at the intersection of AI and process safety is fragmented across at least three distinct dimensions. Firstly, these reviews primarily categorize applications either by their respective process-safety functions (e.g., Hazard Identification (HAZID), monitoring, maintenance) or by the classification of algorithms (e.g., Artificial Neural Networks (ANN), Support Vector Machines (SVM), Convolutional Neural Networks (CNNs)), neglecting to establish explicit connections to varying levels of autonomy and stages of assurance maturity. Secondly, there exists a tendency to delineate "AI for safety" (e.g., AI-enhanced HAZID, monitoring, and dynamic risk assessment) from the "safety of AI systems" (e.g., reliability, cybersecurity, certification), treating these as largely independent subjects rather than recognizing them as interrelated components within the same risk landscape. Thirdly, the majority of surveys culminate in broad recommendations to acquire additional data or develop superior models, yet fail to delineate overarching challenges across the entire safety lifecycle that can inform the prioritization of research endeavors.

This review endeavors to bridge these gaps by employing a cohesive four-dimensional framework that classifies AI contributions according to (i) process-safety function (including hazard identification, monitoring and early warning, barrier and asset management, and learning and governance), (ii) type of intelligence (encompassing descriptive, diagnostic, predictive, and prescriptive), (iii) level of autonomy (ranging from advisory, human-in-the-loop, semi-autonomous, to safety-critical closed-loop), and (iv) assurance maturity (spanning from ad-hoc tools to solutions that can be certified for safety). Within this framework, the study pursues three principal objectives: (i) to systematically categorize AI-enabled safety functions that encompass hazard identification, risk assessment, process monitoring, and risk mitigation; (ii) to critically assess the technological maturity, reliability, and industrial applicability of these AI methodologies within authentic operational contexts; and (iii) to propose a comprehensive, human-centered roadmap aimed at the secure, explainable, and resilient integration of AI into existing process-safety management frameworks. We apply the proposed framework to reinterpret the foundational principles of process safety through an AI-centric perspective (Section 2), to systematically organize AI applications based on intelligence type and autonomy level as opposed to conventional PSM categories (Sections 3-5), to scrutinize the safety of AI itself as an emergent hazard category (Section 6), and to prioritize significant challenges whose resolution would facilitate advancements across various domains of the safety lifecycle (Section 8). In undertaking this approach, the review transcends mere descriptive synthesis and advances a structured, integrative agenda for future research and the industrial implementation of AI-enabled process safety.

2. Process safety foundations in the chemical process industries

The CPIs operate complex, high-hazard systems in which process safety functions as the primary line of defense against low-frequency, high-consequence events. To situate the subsequent discussion of AI-enabled safety systems, it is essential first to consolidate the foundational elements that govern contemporary process safety practice in CPIs. This section, therefore, reviews the core pillars of process safety hazard identification and risk assessment, barrier management and integrity, human and organizational factors, and regulatory frameworks, while highlighting persistent systemic challenges that motivate a transition toward intelligent, data-driven approaches. These foundations subsequently underpin an AI-centric reinterpretation of process safety, providing the conceptual bridge to the taxonomy and applications discussed in later sections.

2.1. Overview and scope of process safety

The domain of process safety within the CPIs constitutes a meticulously structured, system-oriented discipline that aims to avert catastrophic occurrences such as fires, explosions, and toxic emissions. These occurrences, frequently classified as MAHs, are distinct from routine occupational incidents because they are infrequent and entail severe consequences (Tamascelli et al., 2024). Process safety integrates the principles of engineering design, operational governance, human reliability, and management systems to ensure the secure containment of hazardous materials throughout a facility's lifecycle. Historical disasters such as the Bhopal gas tragedy, the Texas City refinery explosion, and the Tianjin port disaster underscore the imperative of comprehensive PSM and the potential societal and environmental repercussions of its inadequacies (Aljarwan et al., 2025). These incidents have precipitated the worldwide adoption of regulatory frameworks that prioritize inherent safety, barrier integrity, and ongoing risk minimization across the design, operation, and decommissioning phases (Cao et al., 2025a).

2.2. Hazard identification and risk assessment

The fundamental principle of PSM is the ability to identify hazards and assess the associated risks systematically. Methodologies such as HAZOP, Failure Mode and Effects Analysis (FMEA), and Fault Tree and Event Tree Analyses constitute the underlying framework for forecasting deviations, identifying root causes, and estimating potential consequences (Ashraf et al., 2022). Although these instruments are well-established, they are predominantly static and heavily reliant on expert interpretation. In contemporary facilities characterized by automation, intricate interdependencies, and rapidly evolving operational conditions, traditional hazard assessments struggle to capture transient dynamics and interactive risks (França and Hollnagel, 2025). Consequently, there is an escalating demand for dynamic, data-driven risk assessment frameworks that can continually refine the risk landscape by leveraging real-time process data. The incorporation of AI methodologies into these conventional frameworks has the potential to enhance

predictive capabilities, facilitating the early detection of anomalous trends and fostering adaptive learning predicated on historical process data (Pu et al., 2023).

2.3. Barrier management and integrity assurance

The discipline of process safety is fundamentally dependent on multiple independent layers of protection that collectively inhibit the escalation from an initiating event to a catastrophic outcome. The LOPA methodology quantifies the reliability of each Independent Protection Layer (IPL), which includes containment systems, pressure-relief devices, alarms, interlocks, and emergency shutdown systems (Elhosary and Moselhi, 2024). It is imperative to maintain the functional integrity of these barriers, necessitating continuous testing, inspection, and verification in accordance with standards such as IEC 61511 for SIS. Nonetheless, post-incident investigations of significant refinery and chemical plant accidents often reveal that barrier degradation, inadequate inspection, and maintenance backlogs are persistent root causes of system failures (Raveendran et al., 2022).

2.4. Human factors and safety culture

The interplay between human and organizational performance is pivotal to process safety. Empirical evidence consistently indicates that a preponderance of process incidents is attributable to human errors or systemic management deficiencies, rather than mechanical failures (Cao et al., 2025b). The intricacies of control-room operations, combined with an excessive number of alarms and ambiguous procedural guidelines, can overwhelm operators, particularly in abnormal or emergency conditions. The field of human-factor engineering seeks to alleviate these challenges by implementing ergonomic interface designs, prioritized alarm systems, and intuitive decision-support frameworks (Silva, 2023).

Equally critical is the development of a robust safety culture, characterized by active leadership involvement, open communication, and a commitment to organizational learning. Training, adherence to procedural guidelines, and reporting of near misses enhance awareness of latent risks and sustain operational vigilance (Kim, 2022). In the era of digital transformation, AI-driven monitoring and simulation technologies can enhance human performance by analyzing operator responses, identifying deviations from established protocols, and providing cognitive support during critical decision-making processes. This synergistic collaboration between human operators and AI may help mitigate human error and enhance overall process safety resilience (Mrugalska and Dovramadjiev, 2022).

2.5. Regulatory frameworks and standardization

The governance of process safety is codified through both international and national standards that delineate requirements about design, operation, and performance. Prominent among these are the IEC 61508 and IEC 61511 standards, which outline a lifecycle approach for safety-instrumented systems; ISO 31000, which outlines risk-management principles; and API RP 754, which defines performance indicators relevant to process safety. Collectively, these frameworks ensure that safety considerations are integrated throughout all phases of plant operation, ranging from conceptual design and commissioning to decommissioning (Orikpete and Ewim, 2024).

In recent years, regulatory authorities have advocated the use of digital technologies to enhance traceability, compliance monitoring, and risk visualization. Digital safety cases, online dashboards, and automated auditing instruments now furnish real-time oversight of safety performance metrics. However, these technological advancements also present novel challenges, particularly regarding data integrity, cybersecurity, and model validation, which necessitate careful consideration to ensure regulatory acceptance of AI-augmented safety systems (Broadribb, 2024).

2.6. Persistent challenges and pathway toward intelligent safety systems

Notwithstanding advances in process-safety methodologies, recurring industrial incidents highlight persistent systemic vulnerabilities, such as disjointed data management, insufficient real-time monitoring, and constrained predictive analytics. Conventional methodologies, while robust for evaluations conducted during the design phase, tend to be reactive within dynamic operational environments. These shortcomings underscore the need for intelligent, adaptive systems capable of continuous learning and informed decision-making.

AI presents a promising avenue to bridge this gap. By leveraging ML, predictive analytics, and digital twin technologies, AI can analyze complex, multidimensional process data, forecast potential hazards, and facilitate timely decision-making for operators. When effectively integrated with existing process-safety standards and human oversight, AI holds the potential to transition process safety from a compliance-oriented endeavor into a proactive, knowledge-driven discipline aimed at averting significant accident hazards (SAHs). The subsequent section examines these emerging AI roles in detail, elucidating their implications for the future of safety management within CPIs (Boudjoghra and Innal, 2023).

Contemporary process plants are transitioning to predictive barrier management, in which sensor data, reliability models, and digital twins are used to evaluate barrier condition in real time. This methodology promotes proactive maintenance and facilitates early intervention, ensuring safety margins are not compromised. The integration of AI into barrier integrity management holds promise for enhancing diagnostic accuracy, optimizing inspection intervals, and quantifying degradation trends with greater precision than traditional methodologies (Adikwu et al., 2024).

2.7. AI-centric reinterpretation of process safety foundations

The fundamental components of process safety, including hazard identification and risk assessment, barrier management, human factors, and regulatory governance, can be reconceptualized through the proposed AI framework, characterized by safety function, intelligence type, autonomy level, and assurance maturity. In the realm of hazard identification and risk assessment, AI initially provides descriptive and diagnostic intelligence through the identification of patterns in historical incidents and unstructured textual data; however, when integrated with real-time process data and digital twin technologies, it progresses towards predictive and, in certain advisory scenarios, prescriptive intelligence pertinent to evolving risk profiles, generally operating within an advisory or human-in-the-loop autonomy framework (Kumar and Sagar, 2025). The management of barriers and assurance of integrity transition from conventional time-based proof testing to predictive, model-driven monitoring, wherein AI reinterprets barriers as assets that are continuously observed and rich in data, rather than as static design assumptions; this shift facilitates semi-autonomous recommendations regarding maintenance and Safety Integrity Level (SIL) revalidation, while still necessitating a high level of assurance maturity for any safety-critical interventions (Arunthavanathan et al., 2024). Human factors and the cultivation of safety culture, which have historically been addressed via training and organizational interventions, can be enhanced through human-centered AI systems that adapt interfaces, alarm systems, and decision-support mechanisms to align with operator workload and cognitive constraints, thereby offering descriptive and diagnostic insights into human performance along with prescriptive guidance under strictly supervised autonomy. Regulatory and standardization frameworks initially designed for deterministic automation must evolve along the assurance-maturity continuum to embrace adaptive learning systems, thereby necessitating a reevaluation of foundational concepts such as SIL, validation processes, and certification protocols for probabilistic,

data-informed models. Collectively, this AI-centric reinterpretation establishes connections between foundational concepts of process safety and explicit categories of intelligence, levels of autonomy, and assurance prerequisites, thereby laying the groundwork for a subsequent taxonomy used to systematically categorize AI applications and analyze how these emergent capabilities are transforming the future of PSM (Rybak and Hassall, 2024).

3. Taxonomy of AI intelligence and autonomy in process safety

Conventional process safety methodologies are more effective when hazards can be characterized by a finite set of scenarios, deterministic models, and a limited set of process variables. Nevertheless, these methods exhibit heightened vulnerability in environments characterised by high-dimensional sensor data, intricate non-linear interactions, and infrequent yet systemic precursors that are not delineated within standard guidewords or fault trees (Azizzadeh Mehmost Olya and Mohebian, 2024). In such scenarios, AI transcends being merely a more efficient execution of pre-existing methodologies; rather, it serves as an indispensable analytical framework capable of revealing latent patterns, adapting to evolving operational parameters, and acquiring knowledge from distributed experiences across multiple facilities. For instance, DL models can identify multivariate anomaly signatures several hours before the activation of threshold-based alarms, while text-mining incident reports can uncover persistent organisational or design deficiencies that remain obscured to solely quantitative methodologies (Mrugalska and Dovramadjiev, 2022).

Instead of replicating conventional PSM components, this section categorizes AI contributions based on (i) the nature of intelligence they deliver descriptive (what is occurring), diagnostic (the rationale behind occurrences), predictive (what is likely to transpire subsequently), and prescriptive (recommendations for action) and (ii) the degree of autonomy they facilitate, ranging from advisory decision support to safety-critical closed-loop control (Liu and Hu, 2022). This classification permits the reinterpretation of established safety functions from an AI-oriented perspective, distinguishing between applications that merely visualise existing risk information and those that fundamentally alter the modalities of hazard detection, assessment, and mitigation. The emphasis here is placed on these dimensions of intelligence and autonomy; subsequent sections of the paper will align them with the process-safety lifecycle, elucidating the deployment of AI capabilities across hazard studies, design and risk analysis, operational monitoring, and organisational learning.

3.1. Integration of artificial intelligence into process safety frameworks

The field of AI is progressively revolutionising safety methodologies within the CPIs by facilitating predictive, data-driven, and adaptive strategies for hazard prevention and mitigation (Ekramipooya et al., 2024). The complex nature of modern process facilities, characterised by extensive interconnectivity, automation, and significant volumes of real-time data, renders traditional safety instruments increasingly ineffective for identifying nuanced anomalies or forecasting emergent hazards. AI offers a cohesive framework for interpreting multidimensional data and generating actionable insights that enhance both preventive and corrective safety protocols.

In this particular framework, AI is optimally understood as a supplementary layer that enhances four core process-safety functions through the integration of various forms of intelligence and degrees of autonomy: (i) hazard identification and knowledge extraction (mainly focused on descriptive and diagnostic intelligence), (ii) dynamic risk assessment and early warning (chiefly reliant on predictive intelligence), (iii) barrier and asset-health optimization (encompassing both predictive and prescriptive intelligence), and (iv) human-AI collaboration and safety governance (characterized by advisory roles and human-in-the-loop autonomy) (Liu and Hu, 2022; Ekramipooya et al., 2024; Aderamo et al., 2024). These functions align with the fundamental aims of process safety: the anticipatory, preventive, detection, and mitigation of MAHs, thereby establishing the structural basis for the taxonomy presented herein.

In Fig. 1, the horizontal axis delineates the progression of artificial intelligence, transitioning from descriptive and diagnostic analytics to predictive and prescriptive decision-making support. Conversely, the vertical axis indicates increasing levels of autonomy, ranging from advisory tools that support human operators to semi-autonomous and safety-critical closed-loop control systems. The quartet of fundamental safety functions, hazard identification and knowledge extraction, dynamic risk assessment and early warning, barrier and asset-health optimization, and human-AI collaboration and safety governance, is methodically illustrated within this framework to clarify how their extent, reactivity, and sophistication amplify in tandem with advancements in both intelligence and autonomy. This conceptual framework elucidates how the integration of AI gradually transforms process safety from a predominantly compliance-oriented, static paradigm into a predictive, adaptive, and continuously evolving risk-governance model.

3.2. Descriptive and diagnostic AI for hazard identification and knowledge extraction

The processes of hazard identification and risk analysis have historically relied on expert-led methodologies, such as HAZOP and What-If analyses, which encapsulate knowledge through systematic dialogue and the enumeration of scenarios. Although these methodologies are effective, they are labor-intensive, executed periodically, and fundamentally qualitative, which limits their adaptability to changing plant conditions. The integration of AI enhances this phase by incorporating descriptive intelligence, synthesizing historical and current events, and diagnostic intelligence, deducing the underlying reasons for those events across extensive and diverse information sources (Macêdo et al., 2022).

Natural language processing (NLP) techniques and text-mining algorithms can autonomously extract causal relationships, failure modes, and event sequences from historical accident documentation, maintenance logs, and safety evaluations. These technological tools transmute unstructured textual information into structured formats, such as ontologies or knowledge graphs, that elucidate the interrelationships among equipment, process variables, and deviations (Gao et al., 2025). Unsupervised and semi-supervised clustering techniques further augment this process by uncovering recurrent patterns in anomalies and near-misses that may remain obscured through manual examination. In more sophisticated applications, hybrid models integrate simulation outputs with these data-driven insights to facilitate dynamic HAZOPs or "living hazard registers" that are updated as operating conditions, process configurations, or organizational practices change.

Within the delineated taxonomy, such applications predominantly furnish descriptive and diagnostic intelligence at various levels of advisory autonomy: they inform and prioritize expert deliberations rather than supplant them. Nonetheless, they significantly transition hazard identification from static documentation towards an ongoing, knowledge-driven paradigm of risk awareness (Chen et al., 2022). Table 1 subsequently revisits these functionalities by explicitly contrasting traditional and AI-enhanced methodologies while delineating their corresponding intelligence and autonomy attributes.

3.3. Predictive AI for dynamic risk assessment and early warning

Traditional risk assessment methodologies, such as QRA and LOPA, are generally based on static failure frequencies and pre-established scenarios that are periodically revised but rarely responsive to real-time fluctuations in plant dynamics. AI introduces predictive intelligence by establishing correlations between streaming process data and evolving risk profiles, thereby enabling a dynamic risk assessment that

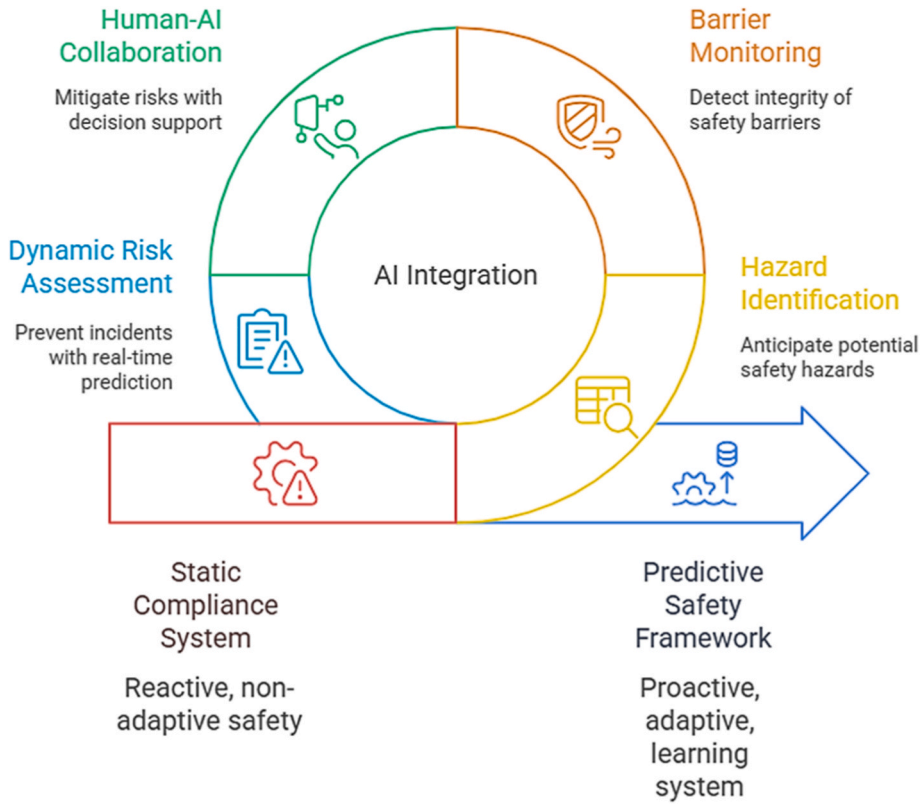

Fig. 1. Conceptual transition from static compliance systems to predictive AI-Integrated process safety framework.

Table 1
Comparison of Traditional vs. AI-Enhanced Process-Safety Approaches.

Safety Function	Traditional Approach	AI-Enhanced Approach	Key Improvements/Benefits	Principal Limitations/ Considerations	Ref
HAZID	Performed through expert-driven HAZOP, What-If, and checklist analyses. Static and periodic, reliant on manual interpretation of process deviations.	NLP and text mining extract hazard information from unstructured reports, maintenance logs, and audit reports. Dynamic updating of hazard registers using live data.	Enables faster and more consistent hazard detection, reduces human subjectivity, and supports continuous "living HAZOPs."	Requires high-quality, standardized text datasets and expert validation of extracted relationships.	Ekramipooya et al. (2023)
Risk Assessment (QRA/LOPA)	QRA based on fixed failure frequencies and historical event data. Updates are infrequent; limited sensitivity to real-time variations.	ML and Bayesian network models correlate live process parameters with incident probabilities, updating risk in real-time.	Provides dynamic risk visualization and adaptive likelihood estimation, allowing for proactive intervention before escalation.	Model bias and data imbalance are prevalent for rare accident events, necessitating ongoing retraining.	Jabłoński and Gucma (2025)
Barrier and Integrity Management	Scheduled inspections and proof-testing following IEC 61511 cycles; failure data updated manually.	Neural networks and probabilistic inference are used to analyze predictive maintenance using vibration, acoustic, and thermal data.	Early detection of barrier degradation, optimized maintenance scheduling, and improved SIL accuracy.	Dependent on sensor reliability and cybersecurity of Industrial IoT networks.	
Process Monitoring and Alarm Management	Threshold-based alarms and control logic; high false-alarm frequency under transient conditions.	DL architectures (CNNs/LSTMs) detect multivariate anomalies and forecast deviations before alarm limits are exceeded.	Reduction in false alarms (up to 40–50 %); enhanced early-warning capability and process stability.	Demands continuous data streaming and robust computational infrastructure.	Sivaraman (2024)
Incident Investigation and Learning	Manual review of reports; qualitative root-cause analysis; limited cross-site learning.	AI-driven pattern recognition and clustering of event databases for root-cause classification and trend identification.	Accelerates learning from near-misses; improves corporate knowledge management and safety benchmarking.	Risk of misclassification if contextual data are incomplete or poorly labeled.	
Human-Machine Decision Support	Operator-centered control; response guided by procedures and experience; limited cognitive assistance.	Reinforcement-learning and XAI systems suggest corrective actions with transparent rationale dashboards.	Shorter response times, reduced cognitive load, and enhanced situational awareness and confidence.	Overreliance on AI advice may lead to automation bias; therefore, mandatory human oversight is essential.	Mietkiewicz et al. (2024)

accurately reflects the current operational envelope rather than relying on averages or nominal conditions (Swadia et al., 2024).

Supervised learning algorithms, including random forests and SVM, can effectively identify precursor states that precede equipment failures or containment breaches by scrutinizing correlations and trends among variables such as temperature, pressure, and flow (Virginia et al., 2025).

Specifically, convolutional and recurrent neural networks are adept at capturing temporal dependencies in sensor data, thereby detecting subtle shifts and multivariate anomaly signatures well before threshold-based alarms are triggered (Aljarwan et al., 2025). Additionally, reinforcement learning further amplifies this predictive dimension by exploring control actions that keep operations within safe bounds amid uncertainty, thereby balancing performance with risk.

Regarding autonomy, most contemporary applications still operate within an advisory or human-in-the-loop framework, generating early warnings, risk assessments, and suggested interventions that require operator validation before implementation (Zamora-Pastor et al., 2025). However, these predictive models also serve as the foundation for more advanced, semi-autonomous control strategies, in which safety-related setpoints or operational modes are adjusted in near-real time. These capabilities underpin predictive safety analytics platforms that issue early-warning alerts, quantify risk probabilities, and propose preemptive measures, thereby transforming risk assessment from a periodic evaluative process into a continuously updated decision-support mechanism (Ad et al., 2025).

3.4. Prescriptive and advisory AI for barrier and asset-health optimization

The dependability of protective barriers encompassing mechanical, electronic, and procedural dimensions remains paramount in averting the exacerbation of perilous incidents. Conventional maintenance methodologies predominantly adhere to time-based schedules, with inspections and proof-testing conducted in accordance with established standards such as IEC 61511, which may result in deterioration going undetected between examination intervals. AI enhances this framework by incorporating prescriptive intelligence advising on optimal interventions and timing, overlaying predictive diagnostics, thereby improving the health of barriers and assets throughout their operational lifespan (Sofoluwe et al., 2024).

ML algorithms, trained on datasets comprising vibration, acoustic, temperature, and process variables, can discern subtle indicators of wear, corrosion, fouling, or calibration drift, thereby facilitating condition-based and predictive maintenance paradigms. Bayesian inference methodologies and probabilistic neural networks provide estimates of time-varying failure probabilities, enabling dynamic revisions to barrier reliability assessments and more precise evaluations of SIL. The concept of digital twins further enriches this discourse by offering virtual representations of critical systems in which degradation phenomena and maintenance protocols can be modeled under various operational contexts.

Within the proposed classification framework, these applications amalgamate predictive intelligence (anticipating barrier degradation and failure probabilities) with prescriptive, advisory autonomy, which recommends maintenance actions, testing intervals, or barrier enhancements while ultimately deferring final determinations to human experts. When integrated into safety-instrumented systems, such AI functionalities must achieve a high assurance maturity to effectively influence safety-critical actuation. Nonetheless, even at the advisory tier, they facilitate earlier identification of barrier deficiencies, optimized resource allocation, and improved transparency regarding barrier health (Shayakhmetova et al., 2025). Table 1 contextualizes these advancements as a transition from static proof-testing to dynamic, AI-enhanced barrier management.

3.5. Human-AI collaboration, autonomy boundaries, and governance

Notwithstanding the rise of automation, human operators maintain an irreplaceable role in overseeing safety-critical scenarios, particularly in the presence of novel or ambiguous circumstances. The incorporation of AI must therefore emphasize collaboration rather than substitution, establishing delineated autonomy boundaries and governance frameworks. In the taxonomy delineated herein, human-AI collaboration predominantly encompasses descriptive and diagnostic dashboards, predictive notifications, and prescriptive advisory systems that function within advisory or human-in-the-loop autonomy tiers (Ramos et al., 2024).

Sophisticated decision-support mechanisms that integrate reinforcement learning with explainable AI (XAI) methodologies can propose corrective measures while offering transparent justifications, such as feature-attribution scores or counterfactual elucidations. AI-enhanced alarm management systems can filter out nuisance alerts, prioritize notifications based on risk significance, and adapt alarm logic to current process conditions, thereby alleviating cognitive overload. At the organizational level, AI-driven safety dashboards consolidate indicators across multiple facilities, thereby aiding leadership in decision-making on resource allocation, regulatory compliance, and cultivating a safety culture (Umoh, 2024).

However, these capabilities introduce novel governance challenges related to model validation, dataset quality, data integrity, cybersecurity, and ethical accountability. Ensuring that AI-generated recommendations are traceable, auditable, and aligned with extant standards is crucial for sustaining trust and mitigating automation bias. Governance frameworks must therefore enshrine requirements for performance monitoring, periodic revalidation, and clearly delineated handover junctures between human and machine decision-making (Alapini, 2025).

Table 1 encapsulates these themes by juxtaposing traditional and AI-enhanced methodologies across pivotal safety functions, highlighting the associated types of intelligence, levels of autonomy, advantages, and limitations.

4. AI applications across the process safety lifecycle

Expanding upon the previously delineated taxonomy of intelligence classifications and levels of autonomy, this section categorizes AI applications according to their respective roles within the process safety lifecycle: (i) hazard identification and knowledge extraction, (ii) quantitative and dynamic risk analysis, (iii) operational monitoring and early warning systems, and (iv) post-event analysis aimed at organizational enhancement. For each phase, it is elucidated whether the underlying models predominantly provide descriptive, diagnostic, predictive, or prescriptive intelligence, and whether they are implemented in advisory, human-in-the-loop, or semi-autonomous frameworks. This lifecycle-oriented perspective mitigates redundancy in methodological details. It underscores how diverse combinations of AI capabilities facilitate barrier management and prevent MAHs across the continuum from design to decommissioning (Tamascelli et al., 2024).

4.1. Overview and rationale

HAZID and risk analysis constitute the theoretical underpinning of process safety, guiding the design and implementation of all preventive and mitigative strategies in the CPIs. Conventional methodologies such as HAZOP, What-If analysis, and QRA remain essential components for ensuring regulatory compliance and demonstrating engineering acumen. Nonetheless, these methodologies are inherently scenario-driven and lack the capacity to integrate high-frequency sensor data, adapt to rapidly evolving operational conditions, or synthesize experiences across geographically disparate facilities (Maduabuchi, 2024).

AI-enhanced applications address these deficiencies by integrating the layers of intelligence and autonomy previously delineated into each lifecycle phase. During the initial design and hazard study phases, descriptive and diagnostic AI systems facilitate automated hazard identification and knowledge extraction. During the detailed risk analysis and operational oversight stages, predictive AI models enable dynamic risk assessments and early-warning mechanisms. In contrast, prescriptive and advisory AI systems optimize barrier integrity and response methodologies. Post-event analysis employs AI to identify

recurring causal patterns and disseminate lessons learned across multiple sites (Stødle et al., 2024). Fig. 2 encapsulates this progression, illustrating how the increasing integration of AI and systemic adaptability transitions the paradigm from static compliance to dynamic, self-learning risk management.

4.2. Automated hazard recognition and knowledge extraction in hazard studies

At the stage of hazard studies, AI applications are built upon the descriptive and diagnostic intelligence previously articulated, yet are now explicitly contextualized as lifecycle tools that inform subsequent design and risk analysis decisions. NLP and text-mining models, encompassing transformer-based architectures such as Bidirectional Encoder Representations from Transformers(BERT) and Generative Pre-trained Transformer (GPT), autonomously extract failure modes, initiating events, and causal relationships from incident reports, audit results, and standard operating procedures (Erfani and Khanjar, 2025).

These models facilitate the development of automated hazard taxonomies and graph-based hazard networks that elucidate the propagation of deviations through units and across sites. By integrating these knowledge frameworks into HAZOP preparations and What-If analysis sessions, engineers can prioritize guidewords, focus on historically underappreciated hazards, and ensure the systematic incorporation of cross-site lessons (Cummings, 2024). Within the proposed taxonomy, such systems predominantly yield descriptive and diagnostic intelligence at advisory levels of autonomy: they pre-populate and continuously enhance hazard registers. At the same time, expert teams retain responsibility for scenario validation and risk prioritization.

4.3. Data-driven and probabilistic risk assessment in design and analysis

In the design and formal risk-analysis phase, AI-driven analytics enhance and expand traditional QRA and LOPA by transforming them into dynamic, data-informed procedures. Algorithms such as random forests, SVMs, and Bayesian networks infer conditional dependencies between critical process variables and event outcomes (Qureshi and Nichols, 2023). When integrated into bow-tie diagrams or fault trees, these models facilitate "living" risk assessments in which event frequencies and barrier reliabilities are systematically recalibrated as new operational data and maintenance records are acquired (Zhou et al., 2024).

DL frameworks, including deep belief networks and CNNs, identify pre-incident indicators in extensive process datasets and forecast runaway reactions, overpressure incidents, or containment failures, with F1-scores exceeding 0.9 in documented applications. Nonetheless, the pronounced class imbalance associated with significant accidents, often constituting less than 0.01 % of all records, demands the implementation of synthetic over-sampling, ensemble learning, or cost-sensitive training methodologies to mitigate overconfidence in predictions of rare events (Peng et al., 2023).

From a lifecycle perspective, these models deliver predictive intelligence that underpins both design determinations (for instance, sizing relief systems and selecting safeguards) and continuous risk assessment. Autonomy predominantly retains an advisory role or incorporates human oversight: engineers interpret AI-derived risk metrics and amalgamate them with expert judgment, regulatory mandates, and financial considerations (Lamina, 2023).

4.4. Operational monitoring, early warning, and barrier management

During the operational phase of the plant, AI applications transition from design-time analysis to real-time monitoring, early warning, and barrier health management, integrating predictive and prescriptive intelligence at higher levels of autonomy. Both online supervised and unsupervised learning models analyze streaming data from DCS and safety-instrumented systems to identify precursor states associated with equipment failures, leakage, or loss of containment (Karishma et al., 2025).

Concurrently, condition-monitoring models process vibration, acoustic, and thermal signals from safety-critical apparatus such as valves, pumps, compressors, and emergency shutdown systems to estimate time-to-failure distributions and residual risk (Sofoluwe et al., 2024). When integrated with digital twins of reactors, storage tanks, and transfer lines, these models facilitate virtual stress testing of barrier performance under hypothetical disturbances, thereby providing a sandbox for evaluating alternative operational strategies without subjecting the physical plant to risk (Bamigbade et al., 2023).

Fig. 2 conceptualizes this progression by positioning AI-enhanced systems along two dimensions: system adaptability (ranging from static to self-learning configurations) and AI integration (extending from basic monitoring overlays to tightly coupled, semi-autonomous control). The four stages, fundamental monitoring, conventional safety evaluation, predictive analytics integration, and real-time risk management, demonstrate how hybrid AI-safety architectures progressively evolve toward adaptive risk governance (Stødle et al., 2024).

Within this phase of the lifecycle, levels of autonomy may range from advisory (e.g., risk dashboards and alarm recommendations) to semi-autonomous (e.g., automatically modifying operating limits or maintenance schedules under established constraints). However, interventions

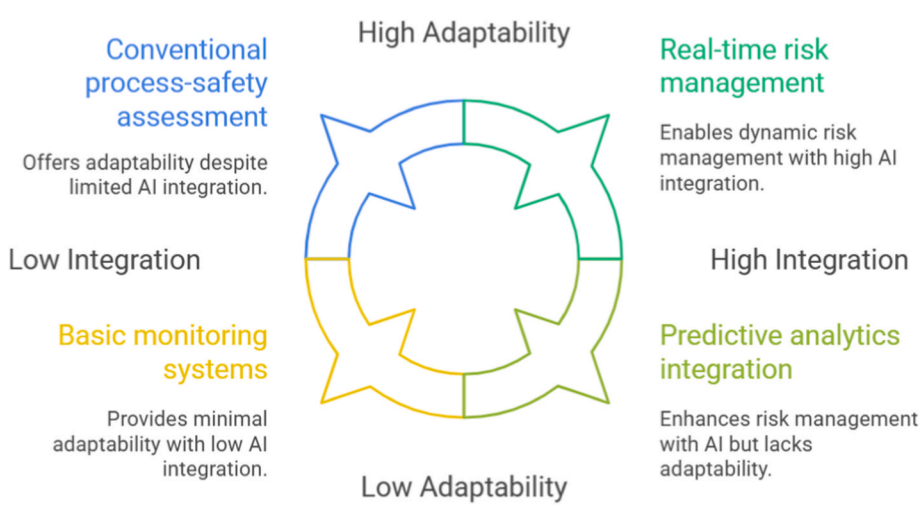

Fig. 2. Evolution matrix of AI integration and adaptability in process-safety frameworks.

that directly activate safety functions still necessitate high assurance maturity and explicit human authorization.

4.5. Post-event learning, emerging frontiers, and research needs

During the learning and enhancement phase, AI applications predominantly focus on systematic learning derived from incidents, near misses, and operational experiences, thereby completing the feedback loop of the process-safety lifecycle. The use of pattern recognition and clustering methodologies applied to extensive event databases enables automatic classification of root causes, trend detection, and benchmarking across multiple sites, significantly reducing the time and resources typically required by conventional manual inquiries. These sophisticated tools furnish descriptive and diagnostic intelligence that underpins revisions to hazard assessments, barrier configurations, and operational protocols.

Emerging research frontiers further amplify this lifecycle integration. Graph neural networks (GNNs) provide more intricate representations of interlinked process units, thereby enhancing the capability to foresee cascading failures and identify network-wide vulnerability hotspots. Moreover, federated learning frameworks enable collaborative model training across multiple locations without exchanging raw data, thereby addressing confidentiality and data governance concerns while simultaneously improving predictive performance (Bamigbade et al., 2023). The integration of AI with high-fidelity digital twins facilitates the simulation of theoretical accident scenarios and the exploration of "what-if" mitigation strategies, thereby fortifying both design and emergency preparedness protocols.

To translate these advancements into robust safety instruments, several challenges must be addressed. Firstly, the transparency and explainability of models are paramount for regulatory endorsement and establishing operator confidence; hence, the incorporation of XAI methodologies into predictive and prescriptive applications is imperative (Kulaklıoğlu, 2024). Secondly, the creation of standardized benchmark datasets and performance metrics is essential for comparative model evaluation and for delineating minimum assurance criteria for varying levels of autonomy. Thirdly, it is crucial to ensure resilience against data scarcity and distribution shifts, particularly for infrequent yet high-consequence events.

In summary, AI facilitates the evolution of HAZID and risk analysis from static, retrospective evaluations to predictive, adaptive risk management that is closely aligned with plant operations and organizational learning processes (Zarei et al., 2022). Table 2 exemplifies this transition by correlating representative AI applications within critical process units with their MAHs, the types of intelligence employed, documented safety benefits, and roles throughout the lifecycle, thereby demonstrating the practical implementation of the proposed taxonomy.

5. AI-enabled monitoring and early-warning systems

AI-enabled surveillance and preemptive alert systems exemplify one of the most sophisticated and practically implementable categories of AI applications in process industries. By building upon the foundational principles of hazard identification, barrier management, and human-machine interaction elaborated upon in preceding sections, these systems aspire to revolutionize monitoring from a reactive approach to alarm management into a proactive framework for risk anticipation. This section integrates methodologies for consolidating advanced anomaly detection models, barrier and equipment health analytics, and human-centered decision-support systems into a cohesive early-warning architecture. A particular focus is on their quantitative advantages, operational ramifications, and the novel dependencies they engender within the overarching process safety paradigm.

5.1. Overview and rationale

Contemporary chemical manufacturing facilities operate in intricate, nonlinear, and closely interconnected environments, where minor deviations can escalate into serious incidents if not identified promptly. Traditional alarm systems are typically predicated on fixed thresholds, simple rules, or single-variable limitations. As manufacturing plants evolve in complexity, these systems tend to produce substantial volumes of false or nuisance alarms while concurrently failing to detect subtle multi-variable patterns that precede critical failures (Mukherjee et al., 2025).

AI fundamentally transforms this paradigm by scrutinizing complex temporal, spatial, and causal interrelations within process data. By analyzing normal operational behavior and its deviations in multivariate time series, AI models can identify weak signals indicative of abnormal operations, assess the rate at which they evolve, and prioritize conditions requiring immediate intervention. When these models are integrated into control and safety frameworks, monitoring transitions progressively from reactive responses to predictive prevention, with early-warning functionalities superimposed upon existing safety-instrumented systems (Niu et al., 2024). The subsequent subsections elucidate the progression of this capability, ranging from anomaly detection to barrier health assessment, equipment prognostics, and human-machine collaboration.

Table 2
Representative AI applications for major accident hazard prevention.

Plant Unit/System	MAHs	AI Technique/Model	Safety Application	Reported Improvement/ Outcome	Ref
Reactor Systems	Runaway reaction leading to explosion or vessel rupture	Long Short-Term Memory (LSTM) and Bayesian Networks	Early detection of exothermic deviations and prediction of overpressure scenarios based on multivariate process data	30–50 % earlier warning of runaway conditions; reduction in unplanned shutdowns	Li et al. (2024)
Storage Tanks (Cryogenic/ Atmospheric)	Overpressure, boil-off gas (BOG) release, or vapor-cloud formation	Ensemble Learning and Digital Twin Models	Predictive simulation of BOG generation and venting control; identification of abnormal temperature-pressure trends	15 % reduction in emission events; improved tank integrity management	Nitin et al. (2023)
Pipeline Networks	Leak, rupture, or ignition along pressurized transfer lines	CNN and Drone-Based Vision Systems	Real-time image and acoustic analysis for leak detection, third-party intrusion, and fire recognition	40 % faster detection of leaks; reduced spill volume and ignition risk	Ekong et al. (2024)
Distillation and Separation Units	Column overpressure or reflux failure leading to release or fire	SVM and Random Forest Models	Process-variable correlation to identify unstable reflux and flooding conditions	25 % improvement in fault prediction accuracy; prevention of overpressure incidents	Taqvi et al. (2022)
Flare and Relief Systems	Flare malfunction or incomplete combustion, causing excess emissions or ignition	DL (Infrared Imaging)	Automated flare-image classification to detect unreported flaring or flame-out events	8 % emission reduction; enhanced environmental compliance	(Al Radi et al., 2024)
Control Rooms/ Operator Interfaces	Human error during abnormal situations leads to the escalation of MAHs	Reinforcement Learning with XAI Visualization	Adaptive advisory systems recommending corrective actions with transparent decision logic	20–30 % faster response time; decreased operator cognitive load	Nitin et al. (2023)

5.2. Process-anomaly detection and prediction

Advanced DL architectures, including CNN and LSTM networks, are increasingly used to analyze multivariate time-series data from DCS and supervisory control and data acquisition (SCADA) platforms. CNNs demonstrate efficacy in capturing spatial dependencies among related variables, such as temperature and pressure profiles across multiple columns or reactors. Conversely, LSTMs are adept at modeling temporal dynamics and long-term dependencies, thereby enabling them to characterize trajectories and identify gradual drifts that may not exceed alarm thresholds for extended periods (Ge et al., 2023).

Empirical studies have shown that LSTM-based anomaly detection systems implemented in refinery environments can reduce false alarms by approximately 45 % and improve the identification of compressor-surge precursors by approximately 30 % compared with traditional statistical monitoring methods (Onyekwe et al., 2023). When amalgamated with DCS platforms, these models furnish a predictive safety layer that evaluates the severity of deviations, remaining time-to-failure, and associated confidence intervals.

XAI methodologies are progressively incorporated into these architectures to enhance trust and diagnostic utility. Variable importance plots, saliency maps, and causal visualizations help engineers understand which measurements triggered an alert and the rationale behind the model's prediction. This degree of transparency is paramount for operator acceptance and for the seamless integration of AI-generated outputs into existing operational protocols. Outputs from anomaly-detection models do not exist in isolation; they directly contribute to the evaluation of safety barriers and equipment health, as delineated in the subsequent section.

5.3. Barrier-health and integrity monitoring

Safety mechanisms, including relief valves, emergency shutdown systems, containment structures, and interlocks, constitute the fundamental elements of protection against escalation following the initiation of a deviation. Conventional maintenance methodologies for these barriers are predominantly time-dependent. Proof tests and inspections are conducted at predetermined intervals, which can overlook degradation that occurs between assessments and may occasionally result in superfluous interventions. The implementation of AI-enabled analytics enables continuous, more selective monitoring of barrier conditions (Yao et al., 2024).

Machine-learning algorithms can be developed utilizing vibration signatures, acoustic emissions, and thermal patterns to identify early indicators of valve leakage, corrosion, or actuator fatigue. For instance, minute alterations in valve-stem vibration may indicate the onset of stiction long before it manifests as a process deviation. Bayesian network inference can subsequently adjust the probability of barrier failure as new information becomes available, rather than assuming failure rates are immutable. These dynamically revised probabilities can be correlated with SIL calculations, thereby ensuring that risk assessments accurately reflect the current condition of protective layers rather than their intended design (Hostens et al., 2024).

Data derived from barrier-health monitoring enhances the process-anomaly detection systems previously discussed. Collectively, they offer a more comprehensive perspective: not only regarding whether the process is transitioning into an unsafe domain, but also concerning whether the safeguards designed to avert an incident are themselves beginning to deteriorate. This integrated perspective naturally extends to broader equipment-health prognostics (Gorjian Jolfaei et al., 2022).

5.4. Equipment-health prognostics and predictive maintenance

Beyond the explicitly safety-classified barriers, AI methodologies are increasingly employed across the wider asset portfolio, encompassing pumps, compressors, turbines, heat exchangers, and rotating machinery, which can impose reliability constraints and pose potential sources of hazardous emissions. Prognostic algorithms leverage historical operational data, event logs, and condition-monitoring signals to approximate the remaining useful life of components and to identify degradation trends under realistic operational profiles (Aliyu et al., 2022).

Hybrid methodologies exhibit particular promise. In these frameworks, physics-based models encapsulate known failure mechanisms, such as fatigue, fouling, and creep, while data-driven models identify plant-specific patterns and deviations from expected behavior. The hybrid model can subsequently estimate degradation rates under varying operational conditions, integrate these into risk metrics, and propose maintenance schedules that minimize both safety risks and economic detriment (Çakır, 2024).

From a process-safety standpoint, equipment-health prognostics help avert hazardous incidents arising from mechanical failures, such as loss of containment due to a compromised seal or compressor damage induced by surges. Concurrently, they furnish structured inputs for human decision-making, which is the focus of the subsequent subsection.

5.5. Human-machine collaboration and decision support

In highly automated industrial facilities, the roles of operators and engineers remain pivotal in detecting, interpreting, and responding to anomalous situations. Nonetheless, these individuals frequently encounter overwhelming alarm signals, intricate graphical user interfaces, and significant time constraints during critical incidents. The implementation of AI-driven decision-support systems aims to alleviate this cognitive strain by integrating information and recommending actions while explicitly maintaining human oversight (Mancuso et al., 2025).

Reinforcement learning agents and adaptive advisory algorithms can be effectively trained in simulated environments to identify control actions that minimize risk and ensure stable operations under disturbances. During actual operational scenarios, these systems present a prioritized array of responses, accompanied by predicted outcomes and confidence metrics, rather than directly altering setpoints or operational modes (Mancuso et al., 2025). When augmented with explainable dashboards, the recommendations are displayed alongside the most significant variables, prevailing trends, and any constraints considered. This facilitates operators' assessments of the reasonableness of a recommendation within the current situational context (Mietkiewicz et al., 2024).

Empirical research suggests that transparent, context-aware AI assistance can significantly reduce response times and minimize human error, particularly during complex operational disturbances (Adusupalli, 2025). Certain systems also monitor workload and fatigue indicators derived from interaction patterns or physiological metrics and adjust the density of information or training prompts as needed. These human-centered functionalities constitute essential foundational elements for the integrated early-warning framework delineated in the subsequent section.

5.6. Integrated early-warning framework and benefits

The amalgamation of anomaly detection, barrier-health analytics, equipment prognostics, and human-AI decision support establishes a comprehensive early-warning and decision-support framework within the safety architecture of chemical process facilities. This framework continuously aggregates and preprocesses data from sensors, control loops, and maintenance systems to enhance real-time situational awareness. Predictive AI models scrutinize these data to detect anomalous patterns, assess evolving risk levels, and prioritize incidents based on likelihood and severity (Aliyu et al., 2022).

The workflow is systematically aligned with the process safety lifecycle. Stage 1 entails continuous operational monitoring and asset

integrity management, in which multivariate data are analyzed to identify early deviations. Stage 2 involves dynamic risk assessment and early-warning functions that use predictive analytics to forecast the potential for escalation, time-to-hazard, and consequence severity. Stage 3 integrates barrier management and human-in-the-loop decision support, incorporating alarm validation, safeguard health assessments, and advisory recommendations while maintaining operator authority over critical safety actions. Stage 4 encompasses governance, organizational learning, and management-of-change processes, ensuring traceability, auditability, and regulatory compliance (Mancuso et al., 2025; Çakır, 2024; Nwulu et al., 2023).

As depicted in Fig. 3, this explicit linkage between stages and the lifecycle reduces nuisance alarms, enhances early detection of weak signals, reduces response times, and strengthens coordination across operations, maintenance, and safety governance, thereby facilitating proactive and resilient process safety management.

When implemented with coherence, such frameworks yield quantifiable benefits, including reductions in false-positive alarms, shorter detection and response intervals, enhanced coordination between operational and maintenance teams, reduced unplanned downtime, lower maintenance expenditures, and improved management of emissions and near-miss incidents (Chen et al., 2024). Furthermore, they enhance the visibility of weak signals and systemic precursors that might otherwise remain obscured.

Concurrently, the increasing dependence on AI within monitoring and early-warning systems engenders a novel category of dependencies. The performance and integrity of AI models themselves become integral components of the overall safety landscape. Consequently, the remainder of this review shifts its focus from AI as a facilitator of monitoring and prediction to AI as a potential source of risk that necessitates rigorous assurance (Pasman et al., 2020).

6. Safety of AI systems in process industries

This section critically analyzes AI not solely as a formidable facilitator of predictive monitoring and decision-making, but also as a novel category of safety-critical elements that require regulation with the same degree of rigor as established protective layers. It integrates emerging viewpoints on the alignment of functional safety, elucidation, cyber-physical systems, and data governance vulnerabilities, as well as on the progression toward safety-certifiable AI within process industries.

6.1. Overview

The integration of AI within CPIs provides substantial capabilities for predictive monitoring, immediate decision support, and adaptive risk management. However, once these models are integrated into control frameworks and safety-instrumented systems, their dependability, resilience, and security issues emerge as significant safety concerns in their own right (Steimers and Schneider, 2022).

In contrast to deterministic automation, AI algorithms operate through probabilistic reasoning and data-centric learning. The training datasets, model architectures, and update strategies influence the model's operational characteristics. This introduces novel failure modes when the models encounter scenarios beyond their training scope or when the data they process is compromised. Phenomena such as model drift, sensor inaccuracies, poor data quality, or intentional tampering can progressively degrade performance without immediate detection, potentially leading to hazardous recommendations or overlooked alerts (Salhab et al., 2024).

Consequently, the imperative to ensure that AI systems maintain safety and trustworthiness transcends peripheral technical matters. It constitutes a fundamental issue in research aimed at loss prevention. The subsequent subsections will elaborate on the alignment of AI with

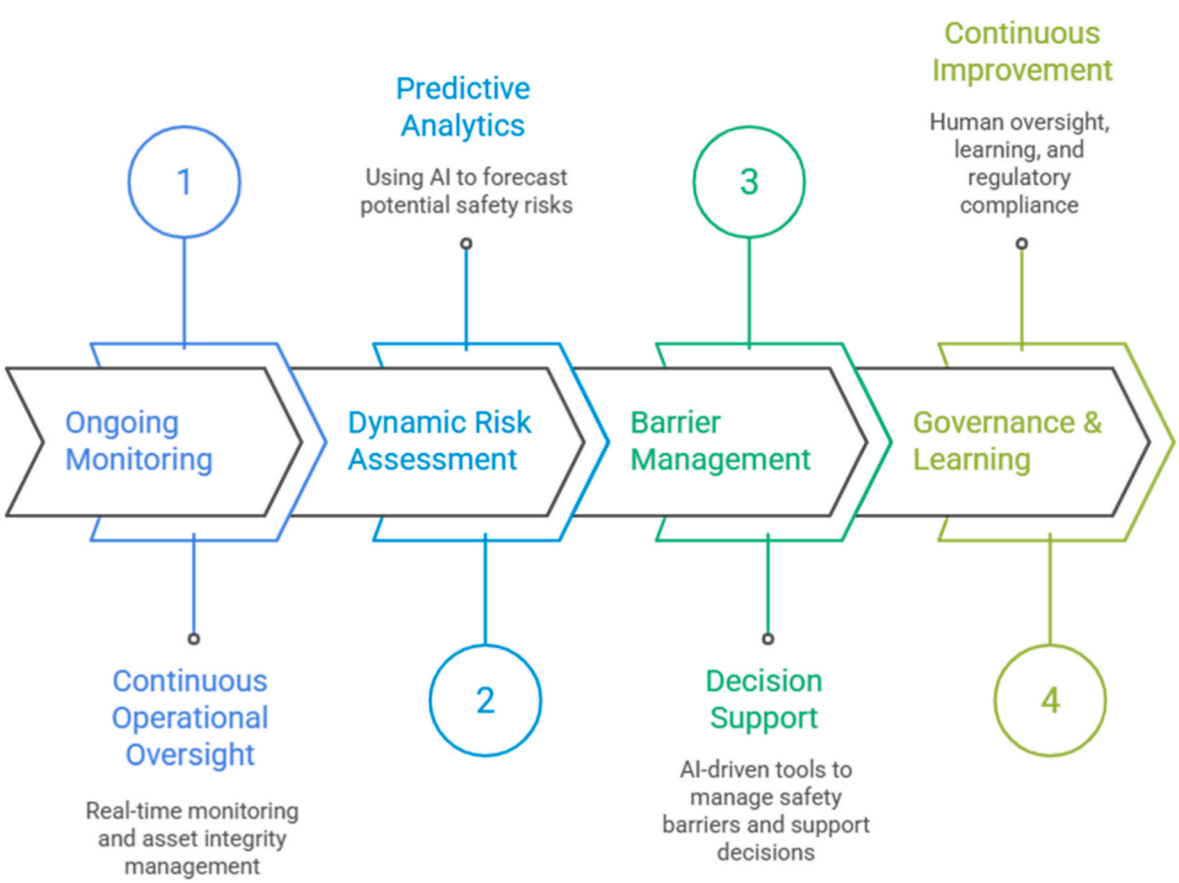

Fig. 3. Sequential workflow of AI-enabled early-warning systems aligned with the process safety lifecycle.

functional safety standards, the approach to explainability, the cyber-physical and data governance risks involved, and the advancements towards safety-certifiable AI.

6.2. Functional safety alignment

AI applications that affect process control or safety-critical functions must comply with functional safety standards, such as IEC 61508 and IEC 61511, which delineate lifecycle requirements for hazard analysis, design, verification, validation, and change management (Aghayev et al., 2024). Although these standards were initially formulated for deterministic logic, their underlying principles remain applicable: safety-critical behaviors must be expressly defined, rigorously tested, and sustained within established limits.

In practice, this often results in hybrid architectural frameworks. AI modules generate advisory or predictive outputs, while deterministic safety Programmable Logic Controllers (PLCs) and hardwired logic maintain ultimate authority over emergency procedures. This delineation ensures that AI can augment situational awareness and enhance decision-making quality without compromising the predictable behavior of safety-instrumented functions (Mietkiewicz et al., 2024). In scenarios where AI is permitted to alter operational thresholds, maintenance priorities, or alarm configurations, its influence is strictly regulated through well-defined envelopes and override protocols.

To substantiate functional safety assertions, AI models necessitate comprehensive documentation. Training datasets, feature engineering decisions, validation metrics, and retraining methodologies must be meticulously recorded to guarantee traceability and reproducibility (Pérez-Cerrolaza et al., 2023). Performance should be characterized not only in average terms but also under worst-case scenarios, and failure modes must be systematically analyzed. The demand for such extensive evidence naturally raises the issue of explainability.

6.3. Explainability and transparency

A primary impediment to the regulatory acceptance of AI in process safety is its inherent lack of transparency. DL algorithms, while highly accurate, often operate as "black boxes." Engineers and auditors may encounter challenges in discerning the rationale for identifying specific anomalies, justifying particular recommendations, or assigning disparate risk scores to seemingly analogous events. This deficiency in interpretability is inconsistent with the level of transparency expected in safety-critical decision-making (Mehendale, 2022).

Methods of XAI seek to mitigate this deficiency. Techniques such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) provide a quantitative assessment of the contributions of individual input variables to a specific prediction. Within the context of process safety, these methodologies can elucidate which factors, such as pressures, temperatures, flows, or indicators of barrier health, inform a risk evaluation or an early-warning signal (Soni, 2024). Visual representations, such as heatmaps superimposed on process schematics or timelines annotated with relevant signals, facilitate root-cause analysis and help operators reconcile AI outputs with their professional understanding.

Enhanced transparency yields two immediate advantages. Firstly, it cultivates trust, thereby increasing the likelihood that operators will engage critically with AI-derived recommendations rather than either disregard them or adopt them uncritically. Secondly, it establishes a foundation for systematic evaluation and auditing, enabling safety professionals to identify systemic biases or mis-specifications within the models (Zarei et al., 2022). Nevertheless, providing explainability alone does not fully address the risks associated with AI; robust cyber-physical and data governance protocols must be implemented in parallel.

6.4. Cyber-physical and data governance risks

The integration of AI with IIoT networks creates novel cyber-physical attack surfaces that extend beyond traditional vulnerabilities in process control. AI systems rely on continuous data streams and model parameters that are frequently updated. Malicious actors who manipulate sensor inputs, intercept communication channels, or modify model weights can induce AI modules to misclassify conditions, suppress alarms, or suggest hazardous actions (Ramapraba et al., 2025).

In response to these threats, AI-driven cyber-anomaly detection systems are being developed to monitor both data patterns and system behaviors. These systems are designed to identify indicators of data poisoning, anomalous access patterns, or abrupt changes in model outputs that are inconsistent with established physical constraints. Concurrently, establishing robust data governance frameworks is imperative to safeguard the integrity, traceability, and accountability of AI systems (Birru, 2025).

Implementing secure data pipelines, model version control, and comprehensive audit trails is critical. These mechanisms enable investigators to reconstruct the specific model version that produced a given recommendation, the data used, and the methods by which that data was collected and validated. Standardized data management procedures across training, validation, deployment, and decommissioning help mitigate the risk of unvalidated model updates or undetected performance degradation (Karishma et al., 2025). Such measures lay the groundwork for more formalized notions of safety-certifiable AI.

6.5. Toward safety-certifiable AI

The overarching objective is to regard AI components utilized in safety-critical applications with a degree of assurance that parallels that of traditional control systems. This necessitates the establishment of more precise safety certification frameworks specifically designed for data-driven models, while remaining consistent with established safety philosophies (Neto et al., 2022). Integral components include quantitative performance objectives, strategies for fault tolerance, robustness assessments amid distributional shifts, and, where feasible, formal or semi-formal verification of essential properties.

Advancements in this domain will hinge on collaborative efforts among standardization organizations, regulatory authorities, industry stakeholders, and academic institutions. The development of shared benchmark datasets, reference architectures, and testing protocols is imperative for comparing methodologies and delineating what constitutes acceptable evidence of AI reliability in safety-related contexts (Mukherjee et al., 2025). Constructs such as assurance cases and runtime monitoring are anticipated to assume significant importance, with AI models operating under continuous performance monitoring and predefined fallback protocols.

Ultimately, AI must be perceived as both a facilitator of enhanced safety and a potential origin of novel risks. Its implementation in high-risk environments must consequently be accompanied by measures of explainability, cybersecurity, robust data governance, and explicit alignment with functional safety standards (Albadr et al., 2025). Only through such comprehensive governance can AI transition from experimental pilot instruments into trusted, safety-critical components of advanced process-safety management systems.

7. Case studies and industrial applications

The preceding sections have delineated the theoretical underpinnings and enabling technologies that enable AI to enhance hazard identification, surveillance, and decision-making within sophisticated PSM frameworks. To substantiate that these advancements extend beyond mere theoretical constructs, this section aggregates representative implementations at both industrial and pilot scales across various domains, including refineries, liquefied natural gas (LNG) facilities,

petrochemical complexes, and chemical storage operations. The chosen case studies underscore measurable improvements in safety metrics, such as reductions in emissions, near-miss incidents, and equipment failures, while concurrently illustrating the necessary organizational and infrastructural changes for successful implementation. In conjunction with the compiled data presented in Table 3, these examples provide an empirical foundation for evaluating the progress, benefits, and ongoing challenges of AI-enhanced safety solutions in the chemical processing sector.

7.1. Refinery and petrochemical plants

Refineries and petrochemical facilities are among the most intricate and high-risk environments within the chemical process sector, making them exemplary contexts for examining and implementing AI-enhanced safety mechanisms. A pilot initiative conducted in 2024 at a Middle Eastern refinery demonstrated the application of DL techniques and infrared imaging technologies for real-time monitoring of flare-gas emissions (Suraparaju et al., 2025). The AI-driven system achieved approximately 92 % accuracy in identifying previously unreported flaring events, thereby facilitating preventive maintenance scheduling and reducing fugitive emissions by nearly 8 %. Beyond mere compliance with environmental regulations, this technological advancement significantly enhanced the facility's ability to detect deviations in combustion efficiency, thereby reducing the risk of overpressure and equipment fatigue associated with flare systems (Bello et al., 2024).

In a distinct initiative at another refinery, a digital twin framework was employed in hydrocracker units to model potential runaway reactions and optimize catalyst replacement intervals. This virtual simulation environment enabled engineers to assess emergency shutdown protocols and analyze operational limits without impeding production activities. The ability to replicate dynamic reaction kinetics across diverse feedstock compositions improved predictive safety margins and reduced the likelihood of thermal runaway, a frequent precursor to disastrous reactor failures. Collectively, these instances demonstrate how AI-facilitated analytics and simulation technologies can transform refineries into adaptive safety ecosystems, where real-time insights drive proactive risk management and asset optimization (Laub et al., 2025).

7.2. Gas and LNG facilities

Gas processing and LNG facilities pose distinct safety challenges attributed to elevated pressures, cryogenic environments, and the potential for rapid phase transitions. AI-driven control methodologies have increasingly been employed to navigate such hazards with enhanced precision. For instance, reinforcement-learning algorithms have been utilized to optimize emergency-shutdown (ESD) protocols within gas-processing installations (Ozowe et al., 2024). By assimilating historical shutdown data alongside simulated disturbance scenarios, the AI system reduced pressure-relief load peaks by approximately 15 % during abnormal operating conditions. This enhancement not only mitigated the risk of cascading overpressure incidents but also prolonged the operational lifespan of safety-relief systems and compressors (Bengherbia et al., 2023).

In applications related to LNG storage, ensemble ML models have been deployed to forecast BOG generation rates, which are crucial for maintaining tank integrity and preventing over-pressurization. These models integrated data on temperature, pressure, and ambient humidity to provide near-real-time prognoses, enabling timely decisions on venting or re-liquefaction (Wu et al., 2024). Such predictive capabilities significantly enhance both operational efficiency and safety assurance, thereby reducing the likelihood of tank rupture or vapor cloud formation. As LNG supply chains expand globally, these AI-driven monitoring systems offer scalable solutions to ensure consistent safety performance across geographically diverse facilities (Włodek and Łaciak, 2023).

7.3. Chemical warehousing and transport

Operations related to the storage, handling, and transportation of chemicals account for a considerable share of process safety incidents, primarily due to human error and noncompliance with established procedures. The implementation of AI-driven computer vision systems has demonstrated substantial efficacy in mitigating these issues. By leveraging drone surveillance and closed-circuit television (CCTV) footage, AI models autonomously detect unsafe loading procedures, improper stacking practices, and violations of personal protective equipment (PPE) protocols in real time (Bahar et al., 2025).

A specific industrial case study conducted in a hazardous materials warehouse found that integrating computer vision into safety inspections reduced manual inspection time by 40 % and near-miss incidents by 25 %. These systems continuously monitor high-risk zones, including loading bays and chemical transfer stations, and generate automated alerts when anomalies are detected. Beyond enhancing immediate safety oversight, the data acquired from these visual analytics is used to analyze behavioral trends, thereby assisting safety managers in identifying specific training requirements and augmenting procedural compliance (Tan and Prayogi, 2025).

By broadening the scope of AI surveillance and analytics to

Table 3
Industrial case studies demonstrating quantifiable safety improvements.

Sector/Facility	AI Application	Process-Safety Objective	Performance Metric	Quantitative Improvement/ Outcome	Ref
Refinery (Middle East, 2024 Pilot)	Deep-learning-based flare-gas monitoring using infrared imaging	Detection of unreported flaring, prevention of overpressure, and incomplete combustion	Accuracy of flare-event detection	92 % identification accuracy; 8 % reduction in total emissions; improved combustion stability	Memarian et al. (2024)
Petrochemical Hydrocracker Unit	Digital-twin simulation for reaction runaway prediction and catalyst-life optimization	Prevention of thermal runaway and reactor overpressure	Catalyst-replacement interval; deviation frequency	20 % extension of safe operation window; fewer emergency shutdowns	Iplik et al. (2022)
Gas-Processing Plant (ESD System)	Reinforcement-learning optimization of ESD sequences	Minimization of pressure-relief load peaks during abnormal conditions	Peak relief-load magnitude	15 % reduction in transient relief-load peaks; improved equipment longevity	(Petsagkourakis and de las Mercedes Puglia, 2022)
LNG Storage Facility	Ensemble-learning prediction of boil-off-gas (BOG) generation	Control of over-pressurization and cryogenic tank integrity	BOG generation rate variance	18 % reduction in abnormal venting; enhanced containment reliability	Widodo and Muharam (2023)
Chemical Warehousing and Distribution	Computer-vision surveillance (CCTV + drone) for unsafe act detection	Real-time identification of PPE violations and improper handling	Near-miss frequency; inspection effort	25 % decrease in near-miss incidents; 40 % reduction in manual inspection hours	Al-Azani et al. (2024)
Integrated Petrochemical Complex	Predictive maintenance and anomaly detection for rotating equipment	Prevention of seal and bearing failures leading to leaks or fires	False-alarm rate; downtime	45 % false-alarm reduction; 12 % lower unplanned downtime	Azim et al. (2023)

encompass warehousing and logistics, organizations are cultivating continuous visibility into safety throughout the chemical supply chain. This comprehensive integration, from production to distribution, represents a notable advancement toward a unified methodology for PSM, supported by digital intelligence (Bhuvaneswari, 2025).

The array of case studies spanning refineries, LNG facilities, and chemical storage operations collectively demonstrates that AI technologies extend beyond theoretical frameworks and are enhancing safety outcomes in real-world industrial contexts. Through applications such as predictive maintenance, anomaly detection, and automated behavioral monitoring, these implementations substantiate AI's capacity to achieve quantifiable reductions in emissions, equipment malfunctions, and human errors (Nadaf, 2024). The increasing adoption of such systems across various process sectors signifies a broader shift toward data-driven, proactive safety management, thereby reinforcing the importance of AI innovation in enhancing loss prevention, operational reliability, and sustainable safety performance within CPIs (Kasimayan, 2025).

Table 4
Technical and organisational limitations in AI-Driven process safety.

Category	Specific Limitation	Underlying Cause/ Description	Implication for Process Safety	Recommended Mitigation/Research Direction	Ref
Data Quality and Availability	Limited access to high-fidelity incident and near-miss data	Proprietary restrictions, inconsistent reporting formats, and underreporting of events	Poor model generalisation and unreliable predictions in safety-critical contexts	Design and evaluate anonymized, standardised cross-industry safety data spaces; quantify the required data volume, diversity, and temporal coverage to achieve specified MAH prediction performance at different intelligence and autonomy levels.	Stefana et al. (2023)
Model Interpretability	Opaque "black-box" behaviour of DL systems	Complex nonlinear relationships are difficult for operators and regulators to interpret.	Reduced operator trust; regulatory resistance to safety certification	Develop XAI methods tailored to continuous process data and safety decisions; experimentally compare explanation formats (e.g., feature importance, counterfactuals, graphical overlays) and measure their effect on operator trust, decision quality, and auditability in simulated and real control-room environments.	Jagannathan et al. (2023)
Data Imbalance for Rare Events	Accident and near-miss data constitute <0.01 % of datasets	Highly skewed distributions bias models toward normal operation	Poor detection of rare but catastrophic incidents	Systematically compare cost-sensitive learning, synthetic rare-event generation, hybrid physics-informed AI, and transfer learning; derive guidelines for when each approach yields reliable MAH detection under extreme imbalance and how uncertainty should be communicated in safety decisions.	Oliveira-Filho et al. (2024)
Cybersecurity and System Integrity	Increased attack surface through IIoT and cloud-based AI systems	Interconnected devices and remote monitoring create vulnerability to cyber intrusions	Potential for simultaneous process and cyber incidents; compromised safety controls	Co-design process and cyber anomaly detection for IIoT-enabled plants; conduct red-team exercises to quantify how combined schemes detect and contain data poisoning, model tampering, and spoofed sensor signals under realistic constraints and latency limitations, aligned with ISA/ IEC 62443 principles.	Pershaanaa et al. (2023)
Human Factors and Automation Bias	Over-reliance on AI outputs; diminished situational awareness	Cognitive dependency on algorithmic recommendations	Misjudgment or delayed intervention during abnormal events	Design and test human-AI collaboration patterns (advisory, veto, shared control) in high-fidelity simulators; measure impacts on workload, situational awareness, and automation bias, and derive interaction guidelines for different autonomy levels and safety functions.	Vicente and Matute (2023)
Standardization and Certification	Absence of formal "AI safety" validation protocols	Existing standards (IEC 61508/61511) do not fully address learning systems, retraining, or drift	Lack of compliance clarity; delayed industrial adoption	Formulate and validate an AI-specific Safety Integrity Level (AI-SIL) concept linking model performance, quantified uncertainty, and retraining policies to PSM decision thresholds; prototype certification workflows for advisory versus closed-loop AI in collaboration with standardisation bodies and regulators.	Iyenghar et al. (2024)
Organisational and Cultural Barriers	Resistance to digital transformation; siloed safety and IT functions	Limited cross-disciplinary communication and trust in AI systems	Slow adoption and fragmented implementation of AI tools	Study multi-site deployment programmes that combine technical pilots with change-management interventions; identify governance models, incentive structures, and training strategies that accelerate safe AI adoption across safety, operations, and IT/OT teams.	Pawar and Shah (2024)
Ethical and Legal Accountability	Ambiguity in liability when AI influences safety decisions	Lack of clear regulatory ownership for algorithmic errors	Legal uncertainty and public distrust	Develop and simulate accountability frameworks (e.g., responsibility matrices, explainable audit trails) for AI-assisted decisions in incident scenarios; analyze how different liability allocations affect organisational behaviour, reporting culture, and public trust in AI-enabled safety systems.	Ajakaye et al. (2024)

Table 3 presents a comprehensive overview of industrial and pilot-scale AI applications in PSM, demonstrating quantifiable reductions in emissions, near misses, and equipment malfunctions across the refinery, petrochemical, LNG, and warehousing sectors.

8. Limitations, research gaps, and future directions

The preceding sections illustrate that AI has the potential to enhance hazard identification, risk assessment, monitoring, and decision-making support throughout the entire process safety lifecycle. Nevertheless, the expansion of these capabilities from pilot initiatives to routine applications within CPIs is hindered by a collection of technical, organizational, and governance deficiencies. These deficiencies encompass data availability, model interpretability, human factors, standardization, cybersecurity, digital twins, and ethical oversight. This section integrates these limitations, conceptualizes them as specific research agendas, and establishes a foundation for a prioritized set of significant challenges and the phased roadmap.

8.1. Data availability and quality constraints

A persistent obstacle for the application of AI in process safety is the inadequacy and fragmentation of high-quality, safety-critical data. Incident reports, maintenance logs, and near-miss documentation are frequently proprietary, inconsistently formatted, and reported incompletely, even within the same organization. Major accidents and serious near misses are inherently infrequent, resulting in most datasets being heavily skewed towards normal operational conditions. Consequently, numerous AI models that demonstrate efficacy in controlled experimental settings fail to generalize effectively to actual plant environments or diverse locations (Blackburn, 2022).

Mitigating this disparity necessitates more than generic appeals for "increased data." It requires establishing structured, cross-industry safety data ecosystems that facilitate anonymized sharing, standardized taxonomies for incidents and near misses, and clearly defined data governance protocols (Patriarca et al., 2021). Table 4 encapsulates this by redefining the “Data Quality and Availability” limitation as a program-level agenda: the design and evaluation of anonymized safety data spaces, as well as the quantification of the requisite volume, diversity, and temporal coverage necessary to fulfill performance criteria for specific MAHs prediction tasks across varying levels of autonomy. In conjunction with the “Data Imbalance for Rare Events” entry, this constitutes a fundamental bottleneck for AI-enabled process safety.

A succinct summary of the primary technical and organizational limitations, their associated implications, and targeted research agendas is presented in Table 4. The final column is intentionally articulated as specific research inquiries or programs to transition from a descriptive gap analysis to actionable future endeavors.

Among these categories, issues about data quality and imbalance, model interpretability, cybersecurity, and standardization emerge as pervasive obstacles. These issues underpin several of the significant challenges elaborated on subsequently and are repeatedly referenced in the roadmap shown in Fig. 4.

8.2. Model interpretability and transparency

The majority of DL models used in process safety operate as opaque “black boxes.” While they may generate precise classifications or predictions, they often fail to elucidate how specific inputs influence a designated output. In contexts where safety is paramount, this lack of transparency is at odds with regulatory requirements for traceability and with the imperative that engineers understand the rationale for classifying a deviation as hazardous before taking action (Aruna et al., 2025).

Consequently, incorporating XAI methodologies, such as SHAP and LIME, into model architecture and implementation is not merely advisable but imperative. These methodologies can illuminate which variables, such as pressures, temperatures, flows, or indicators of barrier health, are instrumental in determining a specific risk score or early-warning signal (Rumapea et al., 2025). Beyond algorithmic development, there is a pressing need for research to examine how different explanation formats affect operator trust, diagnostic precision, and auditability under authentic control-room conditions. As elucidated in Table 4, interpretability extends beyond a mere technical issue and serves as a vital facilitator of forthcoming certification and regulatory endorsement.

8.3. Human factors and cognitive integration

Human expertise remains pivotal to the secure operation of systems, even as AI increasingly supports monitoring and control functions.

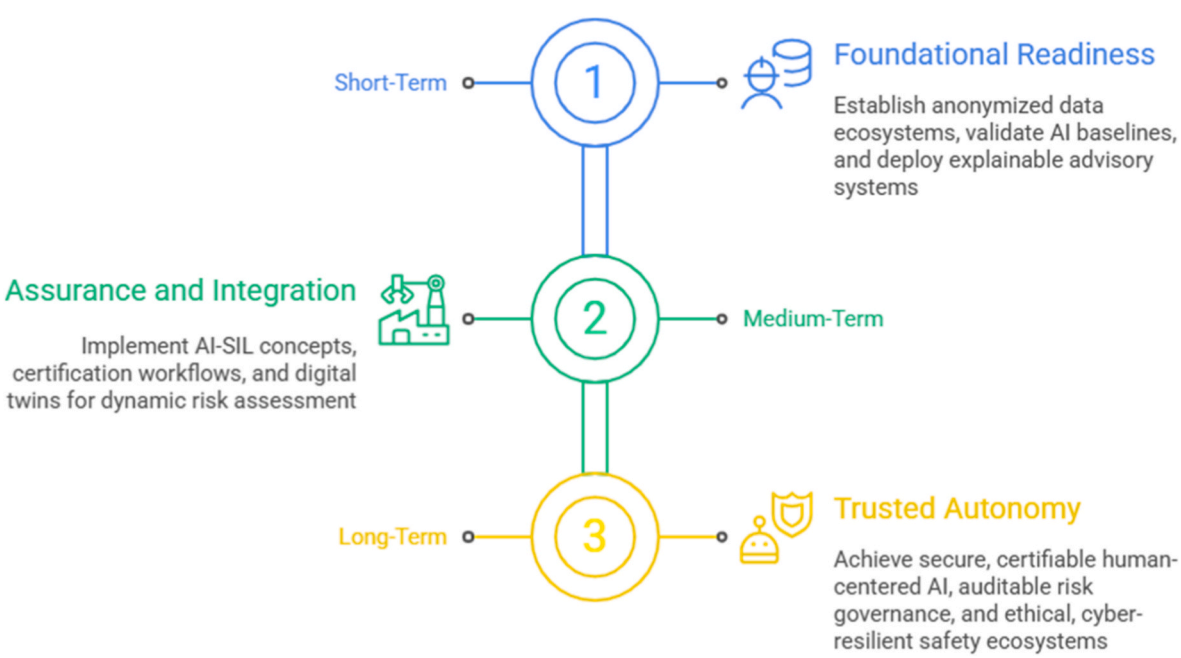

Fig. 4. Phased roadmap for AI integration in process safety.

Nonetheless, inadequately designed AI tools can diminish vigilance and situational awareness. Excessive reliance on algorithmic recommendations may engender automation bias, while convoluted interfaces could inundate operators during atypical scenarios (Hurault et al., 2025).

There is an urgent requirement for research into cognitive ergonomics and adaptive user interfaces that ensure operators remain "in the loop" without experiencing cognitive overload. Promising avenues of inquiry include context-sensitive visualizations that adjust information granularity based on workload, explanatory mechanisms aligned with operator expertise, and training initiatives that use virtual or augmented reality to acclimate personnel to AI-enabled emergency management. Table 4 contextualizes this as a methodical agenda concerning human-AI collaborative dynamics and their quantifiable impacts on workload, situational awareness, and error frequencies. Such investigations will directly contribute to the overarching grand challenge of human-centered autonomy.

8.4. Standardization and certification deficits

At present, no universally endorsed framework defines performance or reliability benchmarks for AI in safety-instrumented systems. Existing standards such as IEC 61508, IEC 61511, and ISO 13849 delineate comprehensive lifecycles for deterministic logic but do not explicitly address learning behavior, model updates, or drift detection (Vesey et al., 2023). This lack of direction complicates the industry's ability to move beyond pilot projects and hinders regulators' capacity to assess AI-based safety claims.

A pivotal research and standardization agenda involves developing an AI-specific Safety Integrity Level (AI-SIL) concept, as indicated in Table 4. Such a concept would align model performance metrics, uncertainty thresholds, retraining triggers, and monitoring requirements with the risk thresholds used in PSM decision-making. Pilot certification workflows for advisory AI (e.g., decision-support dashboards) as opposed to semi-autonomous systems (e.g., AI-influenced setpoint alterations) are required to investigate how these concepts can be assimilated into IEC/ISO lifecycles.

8.5. Cybersecurity and data governance

The convergence of AI with IIoT and cloud-oriented architectures significantly broadens the digital vulnerability landscape beyond conventional control mechanisms. Instances of compromised sensors manipulated data streams, or adversarial modifications to model inputs may result in hazardous control directives or muted alarms (Shah et al., 2025). Consequently, cyber incidents may serve as catalysts or enhancers of operational disturbances.

This necessitates an integrated approach to both process safety and cybersecurity. AI-driven cyber-anomaly detection systems must observe not only network communications but also model outputs and internal states for indications of interference or data corruption. Concurrently, resilient data governance frameworks that incorporate encryption, version control, model lineage, and comprehensive audit trails are imperative to ensure that AI-informed safety decisions are reproducible and legally defensible. Table 4 encapsulates this in the "Cybersecurity and System Integrity" section, highlighting the importance of collaboratively designing cyber and process anomaly-detection mechanisms and evaluating them through pragmatic red-team exercises.

8.6. Digital twins and dynamic risk assessment

Conventional risk assessments are inherently static: they are updated periodically and do not adequately reflect a facility's dynamic condition. Digital twins, which correlate real-time operational data with high-fidelity process simulations, present a viable framework for dynamic risk assessment. When integrated with AI, digital twins can continuously update risk profiles, predict deterioration of barriers, and simulate "what-if" scenarios for emergency management or maintenance planning without subjecting the physical facility to heightened risk (Ogunleye et al., 2025).

Prominent research inquiries include how to ensure sufficient fidelity of digital twins for safety applications, how to synchronize them with live data without destabilizing control mechanisms, and how to incorporate AI-generated risk assessments into existing bow-tie and LOPA frameworks. As delineated in Table 4 and revisited in the grand challenges, the effective amalgamation of AI-enabled digital twins constitutes a fundamental element in transitioning from static risk registries to continuously updated, evidence-based decision-making.

8.7. Ethical governance and cross-sector learning

Ethical and legal considerations about the deployment of AI in Critical Process Industries (CPIs) remain inadequately addressed. When AI impacts safety-related determinations, issues of accountability, liability, and data ethics assume paramount importance. Ambiguity regarding responsibility can erode both internal reporting cultures and public confidence (Ain, 2024).

Regulatory frameworks and organizational governance paradigms must clarify who is accountable when AI-assisted decisions result in incidents, how comprehensible audit trails should be maintained, and how biases in models or data should be identified and mitigated. Cross-sectoral insights from aviation, nuclear energy, and autonomous vehicles are particularly instructive, as these fields have already investigated redundancy, fault tolerance, and human oversight in the context of automation (Neoaz and Amin, 2025). The "Ethical and Legal Accountability" and "Organizational and Cultural Barriers" sections in Table 4 translate these issues into researchable agendas regarding responsibility distribution, governance frameworks, and cultural transformation.

8.8. Grand challenges and prioritized research agenda

A subset of grand challenges identifies systemic bottlenecks that are crucial to advancing AI in process safety. Resolving these challenges is vital for enhancing model performance and ensuring trustworthy deployment of AI in critical safety contexts.

The first grand challenge focuses on creating robust safety data ecosystems for anonymized data sharing, standardised taxonomies, and stringent governance. Fragmented safety data impede effective training and validation of AI models, hindering advances in explainability and risk assessment for rare major hazards (Salhab et al., 2024).

The second grand challenge concerns the development of safety-certifiable AI and AI Safety Integrity Level (AI-SIL) concepts in accordance with IEC standards. The lack of formal assurance pathways confines AI to advisory roles, impeding regulatory acceptance and the transition to safety functions (Pérez-Cerrolaza et al., 2023).

A third grand challenge addresses human-centered autonomy, in which issues such as automation bias and cognitive overload hinder operator effectiveness. Effective AI-assisted decision-making requires thoughtfully designed interaction paradigms and explainable interfaces for safe implementation (Steyvers & Kumar, 2023).

The fourth grand challenge concerns cyber-secure AI architectures and data pipelines. AI safety systems create new vulnerabilities, necessitating integrated cyber-process protections and traceable model lineage to mitigate systemic risks (D'Alessandro and Kirk-Giannini, 2025).

The fifth grand challenge involves incorporating digital-twin-enabled dynamic risk assessment into existing safety workflows. Traditional risk models constrain real-time monitoring, whereas AI-integrated digital twins enable continuous, transparent, and auditable risk assessments (An et al., 2024).

These challenges collectively outline a prioritized research agenda in which advances in data, certification, human-centric design, cybersecurity, and digital twins are essential for developing predictive and trustworthy AI-driven process safety systems. Fig. 4 illustrates the

phased roadmap for AI integration in process safety. The roadmap outlines the progressive incorporation of AI into process safety management. The short-term phase prioritizes foundational readiness, including anonymized safety data ecosystems, baseline AI validation, and deployment of an explainable advisory system. The medium-term phase focuses on assurance and integration through AI-SIL concepts, certification workflows, and AI-driven digital twins for dynamic risk assessment. The long-term phase aims for trusted autonomy, defined by secure, certifiable human-centered AI, auditable risk governance, and ethically aligned, cyber-resilient safety ecosystems.

9. Conclusions

The incorporation of AI into process safety management signifies a transition from traditional, scenario-oriented methodologies to predictive, adaptive, and data-driven safety governance frameworks. This review delineated how descriptive and diagnostic intelligence augment hazard identification and the assimilation of lessons from incidents. In contrast, predictive and prescriptive intelligence facilitate dynamic risk evaluation, barrier and asset integrity monitoring, and human-AI collaborative decision-making within control room environments. The proposed framework establishes correlations among process safety functions, AI capabilities, autonomy levels, and assurance maturity, thereby providing a systematic foundation for aligning AI-enhanced safety measures with sustainability objectives. Trustworthy data ecosystems, safety-certifiable artificial intelligence, and AI-SIL concepts, alongside human-centered autonomy, cyber-resilient architectures, and digital-twin-enabled risk assessments, are recognized as significant challenges whose resolution could directly bolster SDGs 7 (clean energy), 9 (industry, innovation, and infrastructure), 12 (responsible production), and 13 (climate action). For practitioners and regulatory bodies, AI should be considered both a potent enabler and a safety-critical component, necessitating transparent model development, rigorous validation, stringent data governance, and clearly delineated human oversight to achieve safer, lower-emission, and more resilient processes in process industries.

CRediT authorship contribution statement

D. Christopher Selvam: Conceptualization. **T. Raja:** Data curation. **Divyesh Rameshbhai Vaghela:** Investigation. **Mansingh Meena:** Project administration. **Sasmeeta Tripathy:** Resources. **Bhavan Kumar:** Validation. **Honganur Raju Manjunath:** Data curation. **Kulmani Mehar:** Project administration. **Yuvarajan Devarajan:** Writing – review & editing, Supervision.

Declaration of generative AI in scientific writing

The authors acknowledge the use of AI-assisted tools, notably Grammarly, to improve linguistic accuracy, correct grammatical errors, and increase the overall coherence of this article. The content, analytical discussion, and conclusions presented in this work are distinctly the original contributions of the author, with AI tools utilized solely to enhance the presentation and clarity of the text. The use of AI complies with the journal's requirements for transparency and ethical norms in authorship processes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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